

STM32F105 Connectivity & Motor Control Board

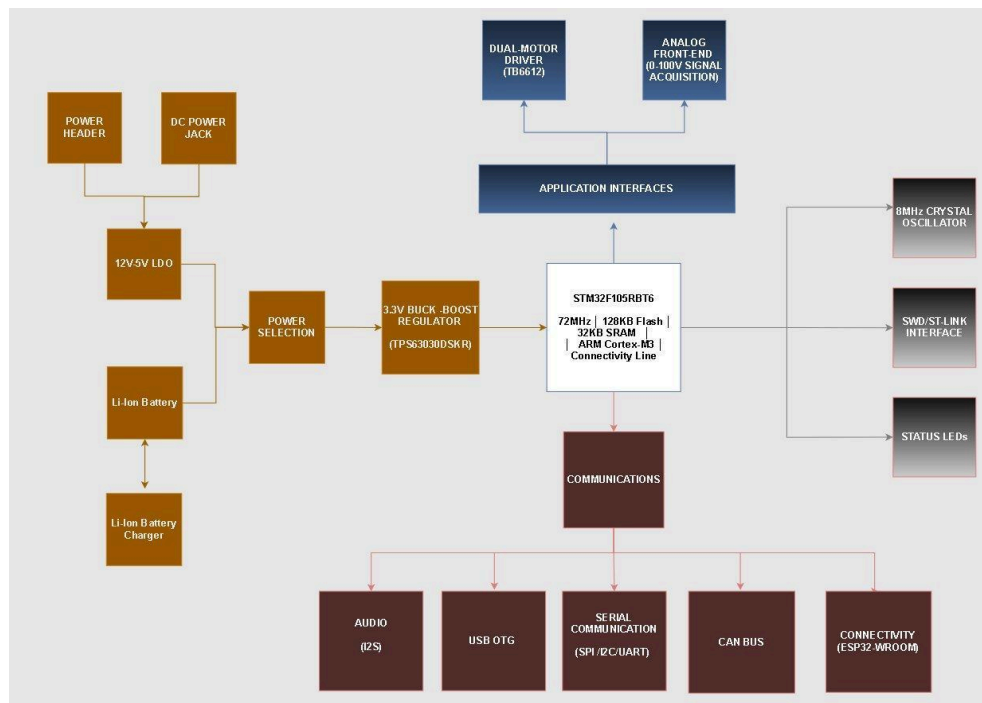
Project Overview

This project documents the design, implementation, and testing of a custom development board centered around the STM32F105RBT6 microcontroller. The goal was to create a versatile hardware platform that bridges the gap between basic microcontroller learning and advanced embedded system design.

The board integrates motor control, high-voltage signal measurement, and robust connectivity options into a compact 4-layer PCB. It is designed to serve as a foundational testbed for developing firmware drivers, testing communication protocols, and validating mixed-signal circuit designs.

System Architecture

The system is built around the STM32F105 MCU, which handles real-time control and data processing. Power is managed through a hybrid input system supporting both wall power and portable battery operation. Specialized subsystems handle motor driving and high-voltage analog acquisition, while standard peripherals provide connectivity to external devices and future wireless expansion.



Detailed Subsystems

1. Core Processing

The heart of the board is the **STM32F105RBT6**, selected for its balance of performance and peripheral set.

- **MCU:** ARM Cortex-M3 core running at 72MHz.
- **Memory:** 128KB Flash for firmware storage and 32KB SRAM for runtime data.
- **Clocking:**
 - **High-Speed:** 8MHz external crystal oscillator for system clock accuracy.
 - **Low-Speed:** 32.768kHz crystal for RTC (Real-Time Clock) functionality.
- **Wireless Expansion:** The PCB includes a dedicated footprint for an **ESP32-C3** module. This is designed as a co-processor setup, communicating via UART. This allows the STM32 to maintain deterministic real-time performance while offloading Wi-Fi and BLE 5.0 stack management to the ESP32 once populated.

2. Power System

Reliable power delivery was a priority, especially given the mixed-signal nature of the board and the potential noise from motor driving.

- **Input Sources:** The board accepts power via a standard **DC Power Jack (12V)** or a **Li-Ion Battery (3.7V)**.
- **Power Selection:** An automatic power selection circuit ensures seamless switching between the DC Jack and the Battery. This prevents back-feeding into the battery when external power is connected and ensures uninterrupted operation during switchover.
- **Charging:** An onboard Li-Ion battery charging circuit manages safe charging profiles when USB or external power is available.
- **Regulation:** A **TPS63030 Buck-Boost Regulator** generates the main 3.3V rail. A buck-boost topology was chosen specifically because the input voltage (from the Li-Ion battery) can vary above and below 3.3V during discharge cycles. This ensures the MCU and peripherals receive a stable voltage regardless of battery state.

3. Analog & Motor Control

This section differentiates the board from standard development kits, adding industrial and measurement capabilities.

- **Motor Driver:** A **TB6612** dual DC motor driver is integrated directly onto the board. It supports up to 1.2A per channel, allowing for direct control of small DC motors without external shields. The STM32 controls this via PWM and GPIO direction lines.
- **High-Voltage Measurement:** A custom **Analog Front-End (AFE)** circuit allows the board to safely measure external signals up to **100V**.
 - *Design Note:* This circuit involves voltage attenuation and protection components to scale high-voltage inputs down to the STM32's ADC reference range (0-3.3V).
 - *Performance:* The signal path is optimized to support the STM32's ADC sampling rates of up to **1MSPS**, enabling capture of fast transient events within the input voltage range.

4. Connectivity & IO

The board exposes the STM32's native connectivity peripherals through standard interfaces.

- **USB OTG:** Full-Speed USB On-The-Go allows the board to act as a device (e.g., CDC/COM port, HID) or host, facilitating easy PC communication for debugging and data logging.
- **CAN Bus:** One independent CAN peripheral is broken out, supporting automotive or industrial networking protocols (CAN 2.0).
- **I2S Audio:** An Inter-IC Sound interface is available for digital audio communication with external codecs or MEMS microphones.
- **GPIO Expansion:** 30 break-out pins provide access to remaining General Purpose IOs, allowing for custom sensor integration or shield stacking.
- **Debug & Status:**
 - **SWD/ST-Link:** Standard Serial Wire Debug header for flashing and debugging.
 - **Status LEDs:** Dedicated GPIO-driven LEDs for visual feedback on power and activity.

Hardware Design Notes

PCB Layout

The board is designed as a **4-Layer Mixed-Signal PCB**.

- **Stackup:** The layer stack is optimized to separate sensitive analog signals from noisy digital and motor drive lines.
 - Layer 1: Signal
 - Layer 2: Dedicated Ground Plane/Layer
 - Layer 3: Power Plane
 - Layer 4: Signal + Power
- **Thermal:** Copper pouring and via stitching are utilized around the voltage regulator and motor driver to assist with heat dissipation.

Assembly

- **Process:** All components were **hand-soldered**. This included fine-pitch components such as the STM32 QFP package and the TPS63030 DSG package.
- **Testing:** Post-assembly testing included continuity checks, power rail verification, and functional testing of individual peripherals (UART, GPIO, ADC).

Technical Specifications

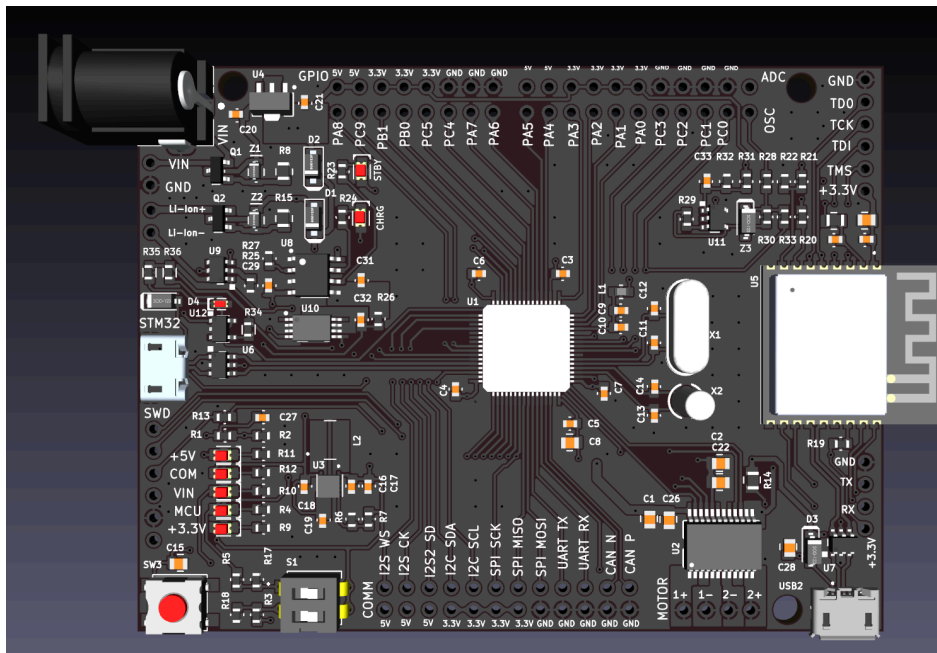
Category	Specification
Microcontroller	STM32F105RBT6 (ARM Cortex-M3)
System Clock	72MHz (via 8MHz External Crystal)
RTC Clock	32.768kHz External Crystal
Power Input	12V DC Jack or 3.7V Li-Ion Battery
Main Rail	3.3V @ High Efficiency (Buck-Boost)
Motor Drive	2x Channels, 1.2A Max (TB6612)
Analog Input	0-100V Range, 1MSPS Capability
Communication	USB OTG, 2x CAN, I2S, SPI, I2C, UART
Wireless	ESP32-C3 Footprint (UART Bridge)
PCB Layers	4-Layer Mixed-Signal

Project Status & Future Work

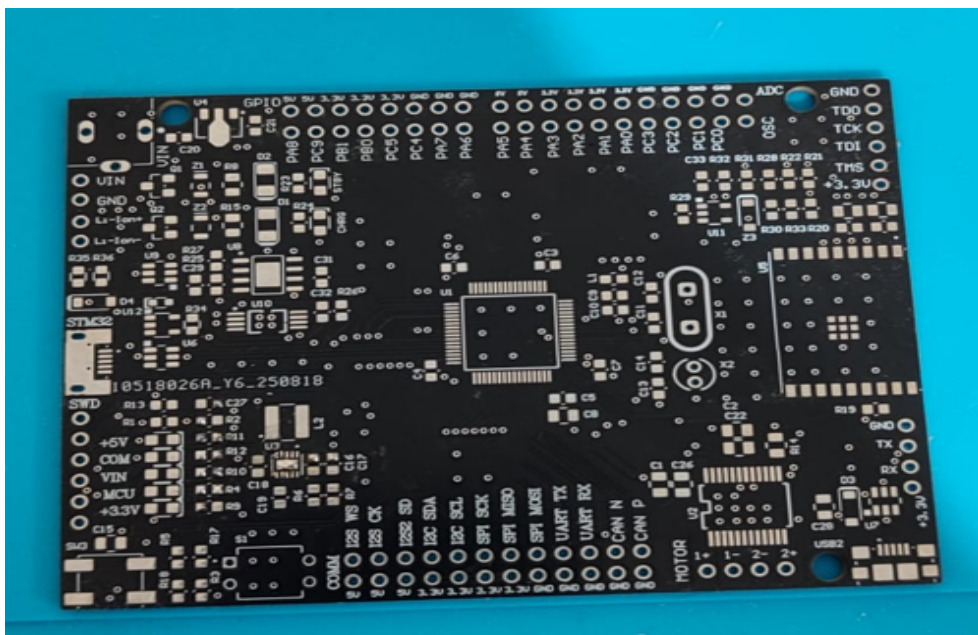
- **Hardware:** Design complete, PCB fabricated, and assembly finished.
- **Firmware:** Basic HAL drivers implemented for GPIO, UART, and ADC.
- **Pending:**
 - ESP32-C3 module population and UART protocol implementation.
 - Full calibration of the 100V Analog Front-End.

Gallery

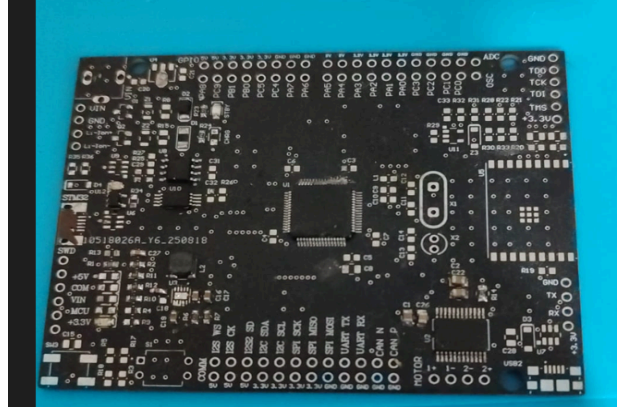
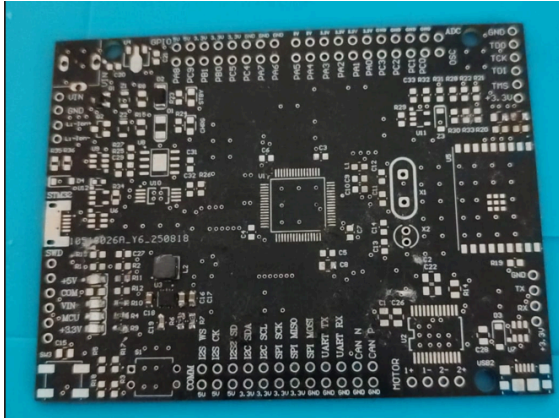
Altium 3D Render:



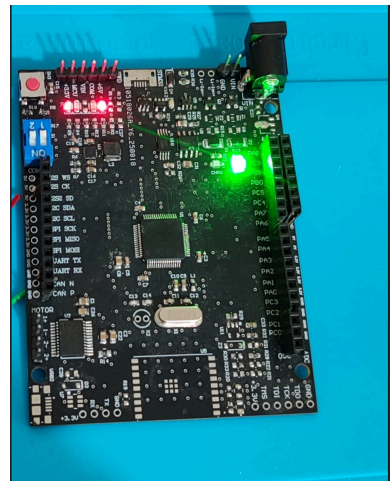
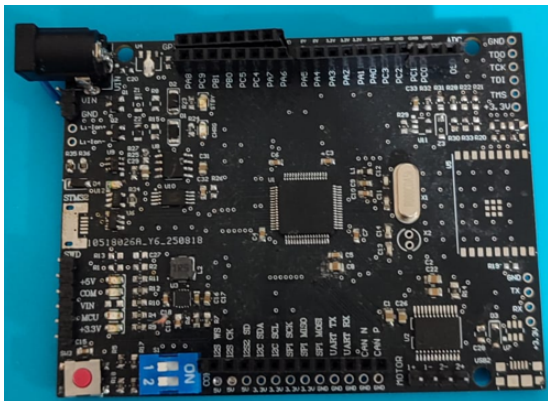
Bare PCB:



Component Soldering:



Latest Board Images:



PC Communication via UART:

